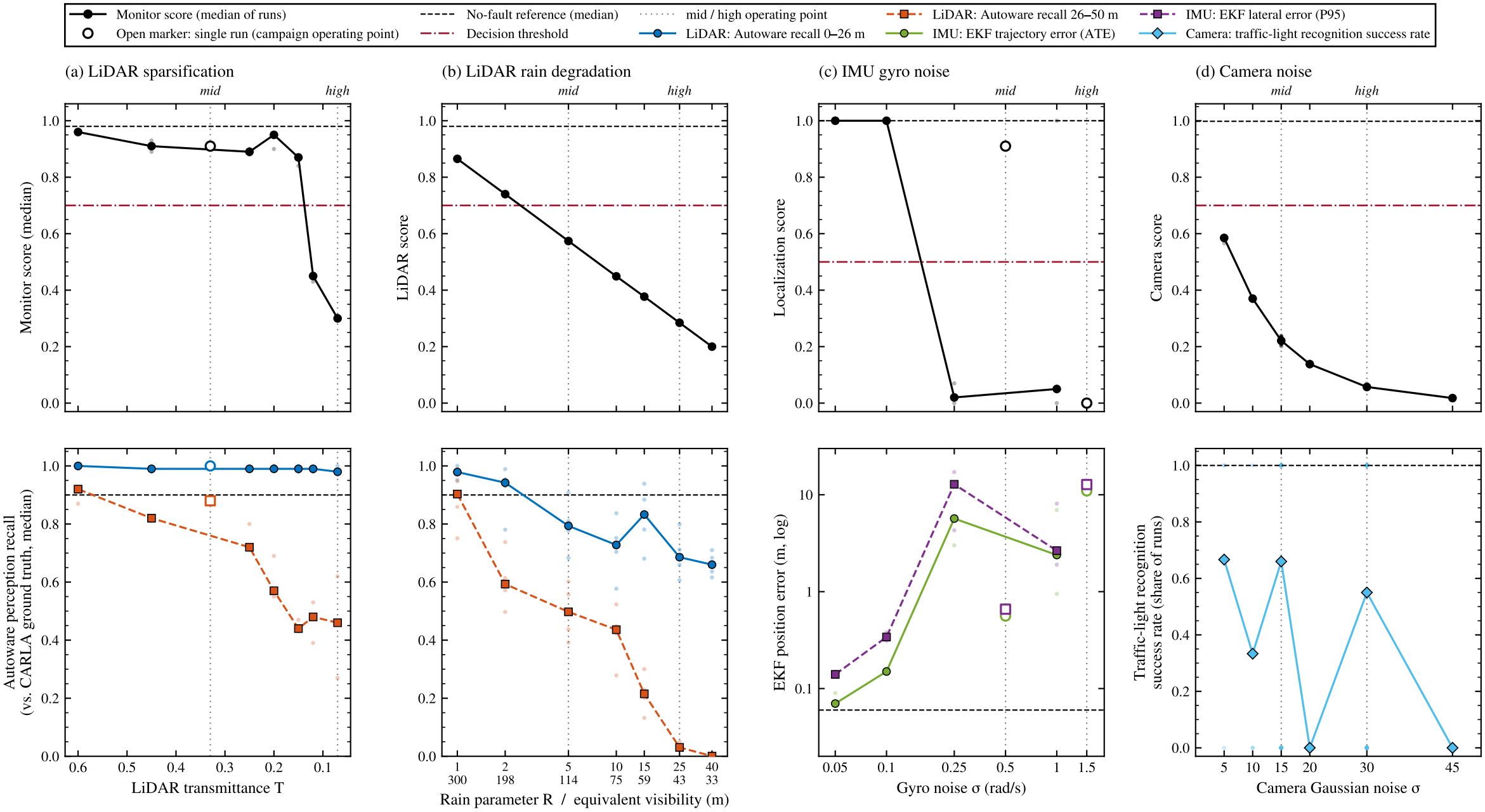

Monitor score (median of runs)
Open marker: single run (campaign operating point)
No-fault reference (median)
Decision threshold
mid / high operating point
LiDAR: Autoware recall 0–26 m
LiDAR: Autoware recall 26–50 m
IMU: EKF trajectory error (ATE)
IMU: EKF lateral error (P95)
Camera: traffic-light recognition success rate
(a) LiDAR sparsification
(b) LiDAR rain degradation
(c) IMU gyro noise
(d) Camera noise
mid
high
Monitor score (median)
LiDAR score
Localization score
Camera score
Autoware perception recall
(vs. CARLA ground truth, median)
EKF position error (m, log)
Traffic-light recognition
success rate (share of runs)
LiDAR transmittance T
Rain parameter R / equivalent visibility (m)
Gyro noise σ (rad/s)
Camera Gaussian noise σ